Experiment Requirements and Repository Audit

Scene graph memory for robotic imitation learning

Prepared for Vincent Pun | September 2026

We do not have all the code needed to run the proposed experiments. The public repository provides a recorded-data perception and memory preprocessing pipeline. It does not provide the downstream policy, training workflow, ablations, or closed-loop evaluation system. Even preprocessing needs correctness checks and repairs before its 3D outputs can support a controlled experiment.

Decision by experiment level

Goal	Ready from this repository	Reason
Generate representations	Partially	Entry point and sample files exist; external models and preprocessing issues remain.
Train one policy	No	No trainable scene-token encoder, diffusion policy, training dataset, or optimizer loop.
Compare the three conditions	No	No baseline or ablation training configurations and no experiment runner.
Measure task success	No	No simulator assets, policy rollout runner, controller integration, or success evaluator.

Scope of this audit

The primary target is the proposal's one-task comparison: full memory, visible objects only, and memory without 3D coordinates. Reproducing every paper result would additionally require the other tasks, RGB and PTP baselines, further ablations, and real-world generalization conditions. These are additional scope, not prerequisites for the smaller project. [P]

Audited revision: 69593b0f886ba6e9f77003176f180a5007bfa0f1. The public API lists one branch, main, with no tags, releases, or issue threads at inspection. The complete tree contains 216 files, including five Python files, five YAML configurations, 40 sample episodes, and two perception-weight pointers. No alternative public release was found. [R]

Evidence strength

- Reviewed all five first-party Python files and all task configurations, traced the entry point, and searched the complete tree for training and evaluation components.
- Downloaded one released trajectory and its depth file, verified their SHA256 hashes against Git LFS pointers, and inspected their schemas and lengths.
- Executed the original NumPy backprojection function in isolation with a synthetic geometric input. Parsed all five Python files successfully.
- Did not install the full perception stack, run neural inference, train a policy, or operate a simulator or robot. Runtime readiness is therefore unverified.

The most consequential new findings are a data alignment defect and an incomplete camera-intrinsics adjustment. Details and exact source locations appear on pages 5 and 6.

Requirements for data and perception

Status means code or data present in the audited revision, not proven execution. External means a separate dependency is documented; partial means important functionality or validation is missing.

ID	Required component	Repository status	Work needed
01	Task specification and protocol	Partial. Task names, classes, and preprocessing YAML exist. No experiment protocol. [C]	Fix task, hypotheses, success rules, splits, seeds, checkpoints, and trial budgets.
02	Demonstration data	Partial. Ten episodes each for cabinet, chicken, giraffe, and pineapple. [R]	Obtain adequate full data and authors' splits; determine exact mapping to paper tasks.
03	Data integrity and access	Partial. Git LFS pointers identify binaries; one trajectory and depth object verified. [D]	Fetch remaining binaries and weights; validate hashes, licensing, complete streams, and storage.
04	RGB depth and calibration loader	Present with defects. HDF5 and MP4 readers plus fixed camera assumptions. [U, H]	Fix source-index alignment; derive camera IDs from metadata; validate timing and transforms.
05	Task object enumeration	Partial. Classes are manually listed; language_instruction is not consumed by Python code. [C]	Use a documented fixed class list or implement the paper's enumeration step if required.
06	Detection and segmentation	Present wrappers; external GroundingDINO and SAM implementations. [T]	Install compatible dependencies and weights; verify masks on actual data.
07	Tracking and late discovery	Partial. CUTIE propagation is implemented; discovery is restricted by configuration. [T, C]	Validate occlusion and reappearance; resolve first-frame-only discovery before generalizing.
08	Object and scene visual features	Present but differs from paper. DINO ViT-B/16 and pooled patch features. [V, H]	Choose release-faithful or paper-faithful features; verify checkpoint loading and shapes.
09	3D centroid extraction	Present with a geometric inconsistency. [H]	Correct the full crop/pad/crop intrinsics; validate units, depth alignment, and extrinsics.
10	Persistent scene memory	Partial. Fixed class slots retain features and centroids; absent boxes are zeroed. [H]	Validate identity assumptions, unobserved slots, invalid depth, and episode resets.

What a verified sample contains

The inspected pineapple trajectory has 152 time steps. It contains six-dimensional Cartesian action arrays, seven-dimensional joint action arrays, gripper commands, robot observations, camera intrinsics and extrinsics, and timestamps. The associated depth file contains 152 frames of 720 by 1280 uint16 depth. This establishes usable source material, but not the intended policy target or complete training data. [D]

The paper's action-space description is not sufficient to choose among these fields without the original trainer and controller convention. Specify the target field, units, absolute or relative semantics, sampling rate, gripper convention, and inverse normalization before implementing a loss.

Requirements for learning and experiments

ID	Required component	Repository status	Work needed
11	Training dataset and action windows	Missing. load_trajectory is a reader, not a training Dataset or DataLoader. [R, U]	Build causal observation/action windows, episode splits, padding masks, and normalization.
12	Scene-token geometry encoders	Missing. No box and centroid MLPs for the downstream policy. [R]	Implement geometry embeddings and combine them with object and scene features.
13	Scene-token transformer and aggregation	Missing. vit.py is the image encoder, not the scene-memory policy encoder. [V]	Implement token roles or positions, masks, attention, aggregation, and conditioning output.
14	Diffusion action policy	Missing. No denoising action model, DDPM loss, or DDIM action sampling. [R]	Obtain authors' implementation or adapt an external diffusion-policy codebase.
15	Optimization and checkpoints	Missing. No optimizer loop, scheduler, validation, policy checkpoint, or resume path. [R]	Implement training, validation selection, saved state, reproducible seeds, and reload checks.
16	Three comparison conditions	Missing. No visible-only or no-3D policy configurations. [R]	Build separately trained conditions and verify all removed input pathways.
17	Original simulation environment	Missing. No task scene assets, robot model, resets, environment runner, or observation adapter. [R]	Obtain original environment or explicitly scope an adapted task and fresh demonstrations.
18	Robot action and controller interface	Missing. Recorded robot data does not supply a deployed control loop. [R, D]	Obtain low-level controller and action conventions; integrate execution and timing.
19	Closed-loop evaluation	Missing. No policy rollout loop, task/subtask checks, or termination protocol. [R]	Implement reset to action to observation loop, success criteria, timeouts, and trial records.
20	Analysis and experiment records	Missing. Mask overlays are not experiment analysis. [T]	Save results by seed and trial; compute uncertainty and comparisons; produce failure analysis.
21	Environment lock and tests	Partial. requirements.txt lists unpinned packages; no test suite or CI found. [R]	Pin tested versions and add critical geometry, synchronization, masking, and reload checks.
22	Compute and storage provisioning	External requirement. Main pipeline explicitly uses CUDA. [H, T]	Secure a Linux NVIDIA host, enough RAM and SSD, and persistent result storage; benchmark demand.

The absence conclusions use both the complete public file tree and source tracing. A class named Transformer inside the visual backbone and perception checkpoint files do not satisfy the missing policy requirements.

What the released code actually executes

Entry point and endpoint

droid_hodor_test_cutie.py parses --config and calls post_process_scene_cutie.main. main constructs HODORProcessor, finds recorded episodes, and calls process_episode. That function reads observations, computes visual and geometric representations, then saves NumPy arrays. There is no action prediction, loss computation, optimizer update, or environment step after that endpoint. [E, H]

File or folder	Actual role	Does not supply
droid_hodor_test_cutie.py	Preprocessing command-line wrapper	Policy test or task-success benchmark
post_process_scene_cutie.py	Image processing, feature extraction, centroids, persistent slots, array export	Trainable scene-token policy encoder or trainer
extract_part_masks_cutie.py	GroundingDINO/SAM wrapper, CUTIE tracking, masks and overlays	General multi-instance graph construction or robot control
droid_utils.py	Read recorded HDF5 and stereo MP4 observations	DROID deployment stack or training dataset
vit.py	Frozen visual feature extraction and checkpoint loading	Scene-memory transformer or diffusion policy
configs/*.yaml	Task classes, slots, detector thresholds, data paths	Learning rates, action horizons, epochs, seeds, or evaluation settings
pretrained/	DINO and GroundingDINO perception checkpoints via LFS	Trained action-policy checkpoints
data/	Recorded demonstrations for four named task folders	Executable simulator or expert demonstration generator

Representation interface that can be reused

For C configured classes and T retained time steps, the main output is episode_pr.npy with shape T by C+1 by 768 and episode_bboxes.npy with shape T by C+1 by 7. These shapes follow from the allocation and save code; the neural outputs were not generated during this audit. Slot zero represents the scene. The seven geometry values are normalized row minimum, row maximum, column minimum, column maximum, followed by x, y, z. [H]

For the pineapple configuration, slots are scene, bowl, and pineapple. Optional outputs include full patch features and a special separated representation for the chicken task. That special separation uses a bounding-box extent heuristic; it is not general instance identity tracking. [C, H]

Feature arrays are written next to the input episode. output.output_dir controls tracker images rather than relocating all generated arrays. Keep raw data immutable in the eventual experiment workspace and direct derived outputs to a separate cache. [H, T]

Preprocessing defects affecting validity

Finding 1 RGB and depth can reference different times

Assessment: Severity: blocking for trustworthy 3D features. Evidence: source trace plus a released trajectory and depth file.

load_trajectory defaults to remove_skipped_steps=True and removes frames when movement_enabled is false. process_episode then enumerates that filtered list and reads depth_frames[frame_num]. It does not preserve the original trajectory index when selecting depth. [U lines 297-304 and 365-375; H lines 424-427 and 439-465]

In the inspected pineapple episode, the trajectory and depth each contain 152 frames. Movement is disabled at original indices 0 through 25, leaving 126 retained steps. The first retained RGB observation is therefore original trajectory index 26, but the caller selects depth index 0. No alternate filtered-depth index or timestamp mapping is present. This finding is based on the released data and code; no detector inference was required. [D]

Required repair: preserve source indices or timestamps, apply the same selection to every stream, and save that mapping with the features and action targets. Then check a paused segment and a continuously moving segment for correspondence.

Finding 2 Backprojection omits part of the crop transform

Assessment: Severity: blocking for the 3D-position comparison. Evidence: original numerical function executed in isolation.

The depth preprocessing crops columns 150:1050, pads 90 rows above and below, then crops rows and columns 50:850. Final pixels correspond to raw coordinates shifted by minus 200 horizontally and plus 40 vertically. For raw camera intrinsics, the transformed principal point must therefore be cx minus 200 and cy plus 40. The current backprojection uses cx minus 150 and unchanged cy. [H lines 194-233 and 387-395]

Synthetic check	Expected	Actual from original function
Raw optical-axis pixel	640, 360	Same input
Processed pixel after specified transforms	440, 400	Same input
Focal length and depth	fx = fy = 500; z = 1 metre	Same input
Backprojected camera point	0, 0, 1 metres	-0.10, +0.08, 1 metres

This is a controlled arithmetic example, not a measurement of robot error. Under these conventions, the camera-plane error is minus 50z/fx in x and plus 40z/fy in y. Its magnitude in real data depends on depth and focal length.

Required repair: derive intrinsics from the complete pixel transformation and verify the extrinsic convention. Add a geometric test and a stationary-object test under camera motion. Confirm the intended calibration with the authors before comparing against their numbers.

Effect on the experiment

Either defect can corrupt the position channel and distort H2. Fixing the policy or adding more demonstrations cannot resolve a systematic input error. Repair and validate the representation before interpreting a no-3D ablation.

Method differences and portability limits

Finding	Source evidence	Implication and required response
Different visual backbone	base.yaml selects dino-vit-b16; vit.py loads DINO ViT-B/16. [C, V]	The paper describes DINOv2. Resolve the target version and document any substitution. [P]
Different global scene feature	HODORProcessor passes global_pool=True; vit.py averages patch tokens and excludes CLS. [H 52-56; V 69-85]	The paper describes a CLS scene feature. Do not describe the released pooled vector as CLS. [P]
Restricted late discovery	Pineapple and chicken disable late detection for every configured class. [C; T 260-264, 307-308]	An object missed initially cannot be newly detected later in those configurations. Test this before using an exploration task.
Fixed semantic class slots	One accepted detection per class; slot mapping requires one slot per class. [T 159; H 74-83, 121-144]	This is not a general graph that adds arbitrary object instances. Multiple same-class objects cannot be represented independently in the main path.
No detector reinitialization after tracking begins	Detected classes only propagate CUTIE and continue, including after an empty mask. [T 283-305]	Recovery depends on the tracker. There is no separate re-detection path to repair an identity failure.
Hard-coded camera identities	load_trajectory overwrites camera_type with fixed serial-number dictionaries. [U 327-355]	New datasets or live cameras require metadata-driven camera selection and explicit calibration.
Invalid geometry can overwrite memory	Centroids start at zero; when no valid depth points exist they remain zero, then overwrite the slot. [H 119-135, 252-253, 314-319]	Distinguish visible-with-invalid-depth from a valid zero coordinate. Retain a valid prior or mark geometry unknown.
No explicit validity or visibility export	Primary export contains features and seven geometry values. [H 327-340, 484-504]	Define seen, visible, and geometry-valid masks for training. Zero values alone conflate several states.
Unpinned and partly unused settings	Unpinned requirements; language_instruction is never read by Python. [R, C]	A configuration label is not proof of functionality. Pin actual dependencies and remove or document unused controls.

These differences do not make every released representation useless. They mean a one-task study needs a declared implementation target, checks for its assumptions, and a clear distinction between reproducing the release and reproducing the paper.

Work needed before running experiments

Obtain the original experiment package first

Send one focused request to the authors. Ask for the release corresponding to the reported experiments and the following materials. This audit has not sent any messages.

- Full demonstrations, episode manifests, train/validation/test splits, and the exact task-folder mapping.
- Scene-token encoder, diffusion policy implementation, training scripts, complete hyperparameters, normalization, action horizons, and sample checkpoints.
- The visible-only and no-3D implementations and their per-seed configurations.
- Original simulator and task assets, robot model, low-level controller, reset logic, observation adapters, rollout runner, and task/subtask success definitions.
- Action field and control conventions, including how Cartesian, joint, and gripper arrays in the data map to commanded actions.
- Clarification of DINO versus DINOv2, pooled patches versus CLS, tracker replacement, frame filtering, and camera crop calibration.
- Reported trial counts, seed counts, checkpoint selection, and a known working software environment.

If the missing code must be reconstructed

Work package	Concrete deliverable	Acceptance check
Repair and validate preprocessing	Index-preserving stream alignment, calibrated centroids, explicit validity masks	Original indices agree across streams; synthetic and camera-motion checks pass.
Build data and model interfaces	Episode splits, causal windows, normalization, geometry embeddings, scene transformer	One batch has documented shapes; input removal checks pass.
Integrate a diffusion policy	Denoising network, training loss, sampling, optimizer, checkpoint save and reload	Tiny-data learning succeeds and decoded actions use correct units.
Implement comparisons	Three separately trained configurations and run manifests	Data and budgets match; the intended information is absent in each ablation.
Integrate task execution	Environment adapter, controller, resets, online memory, success and timeout logic	Expert or supplied checkpoint succeeds; known failure is correctly scored.
Build evaluation and analysis	Raw trials, per-seed summaries, uncertainty, comparisons, failure videos	Metrics regenerate from trial records without choosing results after testing.

Official Diffusion Policy code can supply general infrastructure, but it does not supply this paper's scene-token encoder, task environment, data mapping, or ablation settings automatically. Treat this route as a reimplementation with documented choices. [DP]

Hardware and time consequences

A Linux NVIDIA machine with about 24 GB VRAM, 32-64 GB RAM, and expandable SSD storage remains a reasonable starting allocation, not a measured minimum. Benchmark the repaired preprocessing and a training batch. Simulation additionally needs compatible assets and control code; real-world trials need an available robot, gripper, RGB-D camera, and calibrated controller.

The prior project-guide effort estimate was conditional on accessible infrastructure. It is not a reliable commitment for reconstructing all missing modules and the original environment. Re-estimate after the authors' response and a validated execution route. Offline work alone cannot establish task-success hypotheses.

Evidence references and completion criteria

All source locations below refer to the audited commit. The audit inspected a public release, not private author repositories or unreleased code. A component marked missing was not found in the complete public tree or its listed branches, tags, and releases.

[P]: Paper and supplement

https://arxiv.org/html/2606.01072v2

[R]: Complete repository tree at the audited commit

https://github.com/jianingq/scene-graph-robot/tree/69593b0f886ba6e9f77003176f180a5007bfa0f1

[E]: Preprocessing entry point

https://github.com/jianingq/scene-graph-robot/blob/69593b0f886ba6e9f77003176f180a5007bfa0f1/droid_hodor_test_cutie.py

[H]: Representation processing and export

https://github.com/jianingq/scene-graph-robot/blob/69593b0f886ba6e9f77003176f180a5007bfa0f1/post_process_scene_cutie.py

[T]: Detection and CUTIE tracking

https://github.com/jianingq/scene-graph-robot/blob/69593b0f886ba6e9f77003176f180a5007bfa0f1/extract_part_masks_cutie.py

[U]: Recorded trajectory and camera readers

https://github.com/jianingq/scene-graph-robot/blob/69593b0f886ba6e9f77003176f180a5007bfa0f1/droid_utils.py

[V]: Visual transformer implementation

https://github.com/jianingq/scene-graph-robot/blob/69593b0f886ba6e9f77003176f180a5007bfa0f1/vit.py

[C]: Configuration folder

https://github.com/jianingq/scene-graph-robot/tree/69593b0f886ba6e9f77003176f180a5007bfa0f1/configs

[D]: Verified sample episode

https://github.com/jianingq/scene-graph-robot/tree/69593b0f886ba6e9f77003176f180a5007bfa0f1/data/pineapple/Mon_Sep_22_01%3A28%3A29_2025

[DP]: Official Diffusion Policy repository

https://github.com/real-stanford/diffusion_policy

Evidence retained with the audit

The audit workspace retains the source snapshot, repository metadata and tree, inspected sample schemas, verified trajectory and depth binaries, and check results. The companion requirements CSV provides the 22-row component inventory for tracking status. Full neural execution remains a separate acceptance step.

When we can say we have everything needed

- Data, preprocessing, feature validity, and action conventions are verified.
- Each of the three policy conditions can train, save, reload, and sample actions.
- A compatible environment can reset, execute actions, update memory, and objectively score the task.
- The fixed evaluation protocol produces reproducible per-trial and per-seed records.
- Paper-versus-release differences are resolved or explicitly included in the study's limitations.

Current conclusion: a useful preprocessing starting point is available. A complete, validated experiment stack is not.